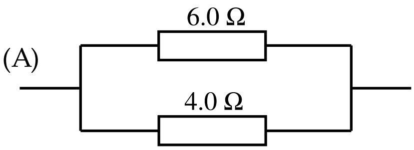
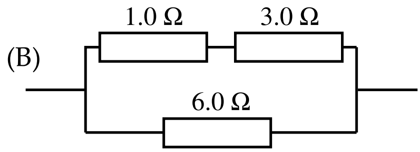
C1 Combinations of Resistors





What is the resistance of labelled combination?

C1.1 a) A b) B

Resistivity

C1.2 Conventional domestic lights are connected with copper cables with a cross sectional area of 1.5 mm^2. Copper has a resistivity of 1.5×10^{-8} Ω m. What is the resistance of 10 m of cable?

C1.3 The two terminals of an 8.00 Ω speaker are connected to two wires in a cable. At the other end of the cable, the wires will be connected to an amplifier. To make the wires look different, one is made of copper, the other is made of aluminium. The wires both have a diameter of 1.2 mm, and the cable is 15 m long. Calculate the total resistance measured between the wires at the end of the cable by the amplifier. Give your answer to three significant figures.
The resistivity of aluminium is 2.5×10^{-8} Ω m.
The resistivity of copper is 1.5×10^{-8} Ω m.

Answers

C1.1

a) $\left(6^{-1} + 4^{-1}\right)^{-1} = 2.4\ \Omega$

b) $\left\{(1 + 3)^{-1} + 6^{-1}\right\}^{-1} = 2.4\ \Omega$

C1.2 $R = \dfrac{\rho L}{A} = \dfrac{1.5 \times 10^{-8} \times 10}{1.5 \times 10^{-6}} = 0.10\ \Omega$

C1.3 $8 + \dfrac{15 \times 2.5 \times 10^{-8}}{\pi\, 0.0006^2} + \dfrac{15 \times 1.5 \times 10^{-8}}{\pi\, 0.0006^2} = 8.5305\ \Omega$